

Simple Probability

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS $P(\text{event}) = \text{favorable outcomes} \div \text{total outcomes}$. A probability is always between 0 and 1 (0 = impossible $\rightarrow$ 1 = certain; $\frac{1}{2}$ = an even chance). Example: a bag holds 5 marbles and 3 are red, so $P(\text{red}) = \frac{3}{5}$.

Round 1 — Retrieve It

1. Write the formula for the probability of an event.

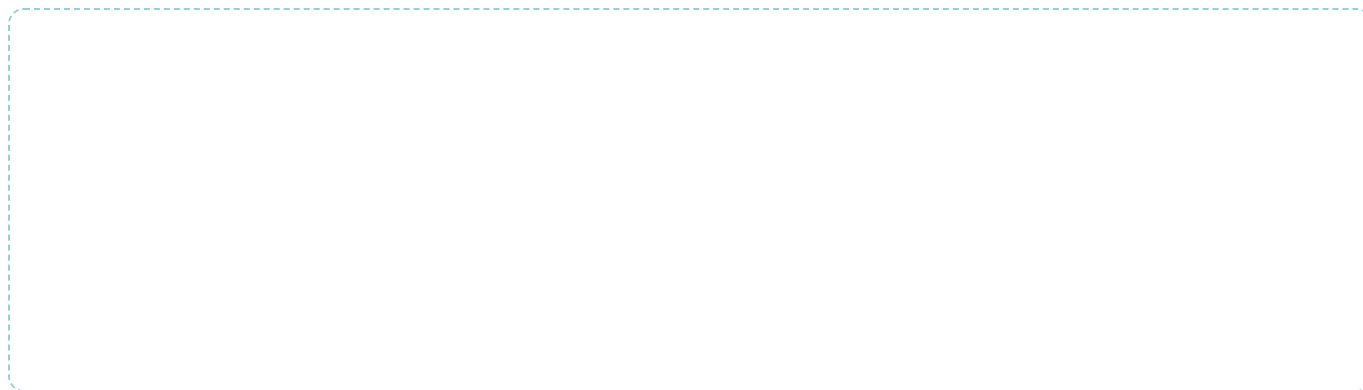
2. What is the smallest and the largest value a probability can be?

3. What does a probability of $\frac{1}{2}$ mean?

4. In a bag of 5 marbles with 3 red, what is $P(\text{red})$ and why?

Round 2 — Sketch It

Draw a number line from 0 to 1. Label 0 as impossible, 1 as certain, and $\frac{1}{2}$ as an even chance. Mark where $P(\text{red}) = \frac{3}{5}$ would sit.



Simple Probability — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. A bag has 5 marbles: 3 red and 2 blue. What is $P(\text{blue})$?

2. You roll a fair six-sided die. What is $P(\text{rolling a 4})$?

3. A spinner has 4 equal sections colored red, blue, green, and yellow. What is $P(\text{green})$?

Round 4 — Reason from Evidence

A coin should land heads about $\frac{1}{2}$ of the time (theoretical). A student flips it 10 times and gets 7 heads (experimental). Explain why the experimental result can differ from the theoretical probability, and what would likely happen with many more flips.

Simple Probability — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how to find the probability of an event. Use one example that is not printed on the cue card.

C**Connect**

Connect experimental and theoretical probability. Describe a trial you could run and how its results would compare to what is possible.

E**Extend**

Create your own bag of marbles. Give the colors and counts, then show how to find the probability of one color as $\text{favorable} \div \text{total}$.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Simple Probability

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. $P(\text{event}) = \text{favorable outcomes} \div \text{total outcomes}$.
2. Smallest = 0 (impossible); largest = 1 (certain).
3. A probability of $\frac{1}{2}$ means an even chance — equally likely to happen or not.
4. $P(\text{red}) = \frac{3}{5}$, because 3 marbles are red out of 5 total (favorable $\div$ total).

Sketch It — look for

A 0-to-1 number line labeled 0 = impossible, 1 = certain, $\frac{1}{2}$ = even chance. $P(\text{red}) = \frac{3}{5} = 0.6$ marked just past the halfway point.

Apply It

1. $P(\text{blue}) = 2 \div 5 = \frac{2}{5}$.
2. $P(4) = 1 \div 6 = \frac{1}{6}$.
3. $P(\text{green}) = 1 \div 4 = \frac{1}{4}$.

Reason from Evidence — look for

Theoretical $P(\text{heads}) = \frac{1}{2}$ comes from what is possible; the experimental result of 7 heads in 10 flips ($\frac{7}{10}$) comes from actual trials and can differ by chance in a small sample. With many more flips, the experimental probability tends to move closer to $\frac{1}{2}$.

ACE — Extend look-for

Many answers possible. Check that the marble counts are given and that the probability is written as favorable $\div$ total for the chosen color.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.